DYLAN VENDITTI

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EDUCATION

Duke University, Durham, NC **Expected Dec 2027**
M.Eng., Robotics and Autonomy
Relevant coursework: Continuum Robotics, Intro to Medical Robotics and Surgical Technologies, Intro to Robotics

Louisiana State University, Baton Rouge, LA **August 2022 – May 2026**
B.S. Biological Engineering, Minor: Robotics Engineering
***GPA:** 4.03 / 4.30 (summa cum laude)*
Relevant coursework: Industrial Robotics, Quantitative Biology in Engineering, Experimental Methods for Engineers

RESEARCH & PROJECT EXPERIENCE

Student Researcher, Autonomous Robotics Systems Sep 2026 – Present
Duke University, Durham, NC
- Adapting a commercial Blue Robotics underwater vehicle for fully autonomous operation and robotic manipulation; evaluating it as a platform for Navy-partnered coral sampling and developing autonomous navigation and control software.

Clinical Biomechanics Research Intern Jan 2025 – June 2026
Ochsner Health, Baton Rouge, LA
- Built Python tooling converting raw motion-capture output into standardized graphical reports, cutting per-patient processing from 2 hours to 10 minutes and eliminating manual transcription error.
- Owned quality control of raw capture data in Vicon Nexus and Visual3D — labeling, gap-filling, and flagging trials compromised by marker dropout or placement error — so instrumentation artifacts were not misinterpreted as real biomechanical changes.
- Analyzed joint kinematics and gait metrics and delivered findings to clinicians.
- Supported data collection on Vicon and Qualisys systems, using hands-on experience with calibration, marker placement, and trial execution to trace downstream data-quality issues back to capture-side causes.

Technical Lead — Capstone: Powered Hip–Knee Orthosis with EMG-Driven Control Aug 2025 – May 2026
Louisiana State University, Baton Rouge, LA
- Led technical direction for a 6-person team building a powered hip–knee orthosis with an actuated knee joint driven by surface EMG.
- Designed the myoelectric control architecture end-to-end: EMG acquisition, intent detection, and mapping intent to velocity commands through a virtual-dynamics model, with a three-layer joint-limit safety envelope (soft-zone deceleration, integrator anti-windup, direct velocity capping) and a feedback watchdog that zeroed output on CAN signal loss.

Industrial Robotics Course Project: 6-DOF Arm Motion Control March 2026 – May 2026
Louisiana State University, Baton Rouge, LA
- Wrote the full motion stack in Python for a low-cost 3D-printed 6-DOF arm to model industrial welding robot
- Wrote DH-parameter-based forward kinematics and damped least squares inverse kinematics with per-joint weighting and servo-limit clamping enforced each iteration.
- Built GUI for path planning, allowing users to input series of waypoints and execute cartesian path between them

ADDITIONAL EXPERIENCE

Design & Project Management Intern — Syracuse Fitness, Syracuse, NY May 2024 – Aug 2024
- Produced dimensioned floor plans and 3D renderings for gyms, universities, and residential facilities; delivered narrated walkthrough presentations supporting approximately $150K in closed contracts.

Undergraduate Teaching Assistant, BE 1251 — Louisiana State University Aug 2024 – Dec 2025
- Coached first-year engineering students through Autodesk Fusion 360, Excel, and core engineering workflows.

TECHNICAL SKILLS

Programming & Analysis: Python (NumPy, PyTorch), MATLAB, C++, Git/GitHub, Microsoft Excel
Robotics & Hardware: ROS 2, surface EMG acquisition/filtering, actuator and motor control, sensor integration, Autodesk Fusion 360, 3D printing
Experimental Methods: Data quality control, protocol design and standardization, system/marker calibration, signal processing, statistical analysis, technical documentation
Motion Capture: Vicon Nexus, Qualisys, Visual3D | Languages: Spanish (elementary)

HONORS & ACTIVITIES

Tau Beta Pi Engineering Honor Society · Volunteers in Public Schools · IM League Basketball Captain